



Transient coupled thermo-elasticity analysis of a temperature-dependent thick-walled cylinder under cyclic thermo-mechanical loads

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Abstract

Several studies have reported the solution of the classical coupled thermo-elasticity for thick-walled cylinders under different boundary conditions; however very limited studies have been conducted on the analysis of temperature-dependent thick-walled cylinders under cyclic nonlinear boundary conditions. Therefore, this study investigates the transient response of temperature-dependent thick-walled cylinders under cyclic nonlinear thermo-mechanical loads based on classical coupled thermo-elasticity. Also, it studies the impact of considering temperature-dependent material properties (TDMP) and temperature-independent material properties (TIMP) hypotheses on the accuracy of the results. The governing equations of the classical nonlinear coupled thermo-elasticity are numerically solved based on the finite-difference method, namely the Crank-Nicolson method. Then, an experimental setup is designed to further investigate the impact of considering TDMP and TIMP through a temperature measurement of the outer surface of a machine gun barrel under burst shooting. It has been found that considering TIMP overestimates the residual temperature and residual stress values through the cylinder thickness compared with TDMP. Moreover, based on the temperature measurement, TDMP estimates results with higher accuracy of 7% than TIMP.

Article Highlights

- Fully coupled-thermoelastic analysis of a thick-wall cylinder consider temperature-dependent material properties.
- Investigation on the effect of thermal accumulation due to cyclic thermo-mechanical loads.
- A comparative study between the impact of considering TIMP and TDMP on the accuracy of the results.

Keywords Classical coupled thermo-elasticity · Cyclic thermo-mechanical shock · Temperature-dependent material properties · Nonlinear boundary conditions

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1 Introduction

Thick-walled cylinders are widespread components utilized in several industrial fields such as mechanical, aerospace, naval and offshore, petrochemical plants, military, and electronics. They can work as pressure vessels in different structures under different mechanical or thermo-mechanical loading conditions.

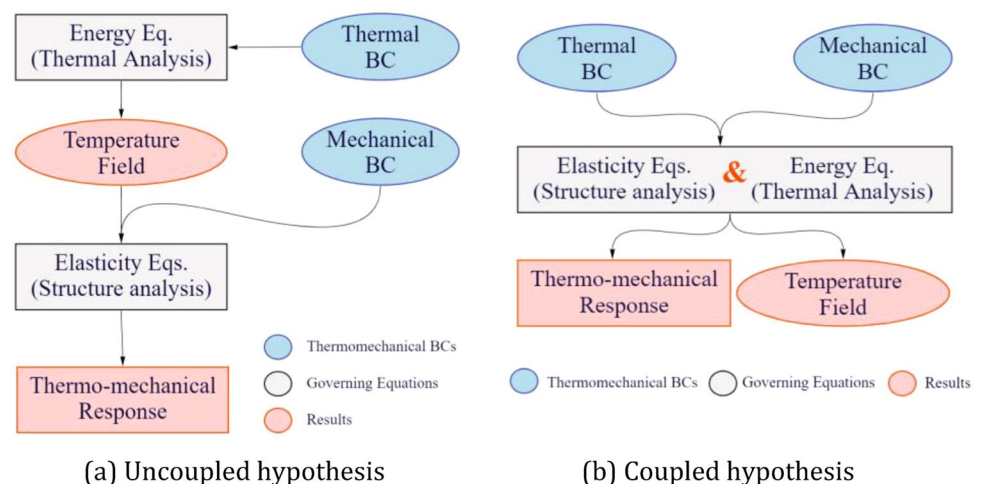
Some of these structures work under timely-repeated thermo-mechanical loading conditions, as in machine gun barrels that are subjected to continuous shooting. Modeling of the effect of these cyclic thermo-mechanical loading conditions is difficult and maybe a challenging affair. This is because the time between two successive input loads is generally too short to allow the cooling down of the cylinder. Therefore, this leads to thermal accumulation in the cylinder material, which causes variations in stress and temperature distributions through the cylinder thickness [1, 2]. In addition, heat accumulation in cylinders may cause remarkable variations in their mechanical and thermal properties. Therefore, the design process of these timely-repeated thermo-mechanical loaded structures is a thriving field of study because it involves coupled multi-physics design problem that incorporates a nonlinear material response due to mechanical and thermal loading.

Several published work investigate thick-walled cylinder response due to thermo-mechanical boundary conditions based on thermo-elasticity [3–6]. Thermo-elasticity is a mathematical model that predicts the structure response due to thermo-mechanical loading conditions. The main governing equations of thermo-elasticity are nearly the same as those for elasticity equations besides the energy equation [7]. According to the studies reported, the solution of thermo-elasticity can be arranged under two groups of mathematical models, namely: uncoupled and

coupled thermo-elasticity models [8, 9]. The main difference between the two groups is the coupling between the displacement and temperature fields in the governing equations of thermo-elasticity. Unlike the coupled thermo-elasticity, the displacement field on the energy equation is ignored in uncoupled model.

Uncoupled thermo-elastic models can be classified as static and dynamic models. The difference between the two groups is the consideration of inertia effects, in the elasticity equations, that are accounted for only in dynamic models. However, including the inertia effect is thought to be necessary, especially in rapidly time-dependent-loaded structures. It is worth noting that, as the displacement response is ignored in the energy equation, the energy equation can be solved firstly and separately from the elasticity equations. Then, the temperature field can be fed to the model of elasticity in order to estimate the thermo-mechanical response of the structure, as shown in Fig. 1(a). Several reported studies discuss the solution of thermo-elasticity based on uncoupled models [10–16]. Sentürk et al. [10] used the static uncoupled model to analytically investigate the thermo-mechanical response of a machine gun barrel. They solved the steady state heat conduction equation to get the temperature distribution through the barrel thickness. Then they calculated the stresses based on the temperature results and elasticity theory ignoring the inertia effect on the equation of motion. On the other hand, Farhan et al. [11] considered the inertia effect on the elasticity model, and then they numerically investigated the stresses and temperature distribution through a cylinder thickness based on the dynamic uncoupled model approach. They considered the numerical approach based on the Finite-Difference FD rather than the analytical model. As they considered in their study temperature-dependent material properties

Fig. 1 Flow work solution of thermo-elasticity problem



(TDMP); then, the analytical solution in that case become more complicated.

Although uncoupled models give an accurate response prediction for several thermo-mechanical loaded structures, they failed to predict actual physical behaviors of structures worked under sudden thermo-mechanical loading conditions (in which the loading period has the same order as the structure's lowest natural frequency magnitude [7]). In this case, the use of coupled models in which elasticity and energy equations are coupled through both displacement and temperature fields is thought to be a must. The thermo-mechanical response in coupled models is estimated through the simultaneous solution of the elasticity and energy equations, as shown in Fig. 1(b).

Coupled thermo-elasticity models were used in some previous studies to evaluate the thermo-elastic responses of thick-walled cylinders under different boundary conditions. These studies can be categorized based on the solutions to analytical and numerical solutions. On the one hand, the analytical solutions are based on fundamental mathematics, as in [17–20]. Eslami et al. [17] presented a closed-form solution for the classical coupled thermo-elasticity problem in cylindrical coordinates; the solution is based on the Eigen function of Fourier expansion. Using the same concept, Kouchakzadeh et al. [19] investigated a closed form solution of a rotating disk (the body force is included in the elasticity model) based on the coupled thermo-elasticity.

On the other hand, numerical solutions based on FD or Finite-Element FE are also used for coupled thermo-elasticity problems [21–23]. They provide more powerful tools as they can account for more complicated loading conditions and nonlinear material properties, which is challenging in analytical solutions. Abd-Alla et al. [23] presented an FD solution for a coupled thermoelasticity problem in cylindrical coordinates considering nonlinear boundary conditions. The FD model is based on the unconditionally stable Crank-Nicolson approach. All aforementioned studies [17, 23] are built under the assumption of temperature-independent material properties (TIMP). However, few studies consider temperature-dependent material properties (TDMP) to investigate the solution of coupled thermo-elasticity for simple loading conditions [24, 25].

The difficulties of coupled models in comparison with uncoupled models can be summarized as (i) the need to include cyclic nonlinear thermo-mechanical working conditions as it occurs in real systems. These needs deep investigation to clearly depict the effect of thermal accumulation on stress and temperature responses due to highly cyclic loading conditions, and (ii) considering TDMP in elasticity and energy equations needs more investigation to depict its effect on calculations accuracy. Also, it is interesting to mention that considering TDMP

hypothesis converts the problem to a nonlinear thermo-elasticity problem. These types of problems are mentioned as challenges in previous studies [26]. Overcoming these challenges on coupled thermo-elasticity approach leads to accurately simulating thick-walled cylinder response under different working boundary conditions.

This paper presents a coupled approach to investigate the transient thermo-mechanical response of temperature-dependent thick-walled cylinders under highly cyclic nonlinear thermo-mechanical shocks. A detailed numerical solution procedure is implemented to solve the nonlinear coupled thermo-elasticity governing equations. The developed numerical model is based on the Crank-Nicolson approach. Then, the model results are validated with the results from the literature. Moreover, a comparative study is performed to investigate the impact of considering TDMP and TIMP hypotheses on the accuracy of the results. Furthermore, an experimental setup is designed to further investigate the impact of considering TDMP.

The paper is structured as follows, the main governing dynamic equations that address the classical nonlinear coupled thermo-elasticity in cylindrical coordinates are formulated in Sect. 2. Then, Sect. 3 presents the detailed finite-difference numerical solution of the nonlinear coupled thermo-elasticity for general thermal and mechanical boundary conditions. Section 4 presents the model validation with previous work in the literature, followed by the experimental setup and the comparative study of considering TIMP and TDMP on the accuracy of the results. Finally, conclusions are drawn in Sect. 5.

2 Governing equations

Consider a thick-walled cylinder made of isotropic material with an inner radius R_i and outer radius R_o (as shown in Fig. 2) subjected to axisymmetric thermo-mechanical

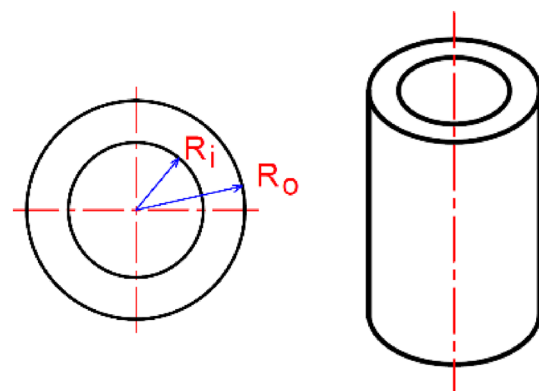


Fig. 2 Schematic drawing of a thick-walled cylinder

loading conditions. The cylindrical coordinates (r, θ, z) and axisymmetric plane strain conditions have been assumed; thus, $u_\theta = u_z = 0$ and $u_r = u(r, t)$. Therefore, the strain–displacement relations in cylindrical coordinates can be described as the following:

$$\varepsilon_r = \frac{\partial u_r}{\partial r}, \varepsilon_\theta = \frac{u_r}{r}, \varepsilon_z = \gamma_{r\theta} = \gamma_{rz} = \gamma_{\theta z} = 0 \quad (1)$$

The classical coupled approach of the thermo-elasticity theory is used to investigate the thermo-mechanical response of the thick-walled cylinder due to its loading conditions. The main governing equations of this approach are the elasticity equations and the thermal energy transfer. On the one hand, the main governing equations for the elasticity theory are [27]:

(a) The constitutive relations for isotropic material:

$$\begin{aligned} \sigma_r &= \frac{E}{(1+\nu)(1-2\nu)} [(1-\nu)\varepsilon_r + \nu\varepsilon_\theta] - \frac{E\alpha}{(1-2\nu)} T \\ \sigma_\theta &= \frac{E}{(1+\nu)(1-2\nu)} [\nu\varepsilon_r + (1-\nu)\varepsilon_\theta] - \frac{E\alpha}{(1-2\nu)} T \end{aligned} \quad (2)$$

(b) The equation of motion “due to the absence of shear stresses and ignoring the body forces”:

$$\frac{\partial \sigma_r}{\partial r} + \frac{\sigma_r - \sigma_\theta}{r} = \rho \ddot{u}_r \quad (3)$$

Substituting Eqs. (1) and (2) into Eq. (3) to put the equation of motion in terms of radial displacement components as:

$$\frac{\partial^2 u_r}{\partial r^2} + \frac{1}{r} \frac{\partial u_r}{\partial r} - \frac{u_r}{r^2} - \frac{\alpha(1+\nu)}{(1-\nu)} \frac{\partial T}{\partial r} - \frac{\rho(1+\nu)(1-2\nu)}{E(1-\nu)} \ddot{u}_r = 0 \quad (4)$$

where $\varepsilon_r, \varepsilon_\theta, \varepsilon_z, \gamma_{ij} (i, j = \{r, \theta, z\}; i \neq j)$, $\sigma_r, \sigma_\theta, E, \nu, \alpha, T, \ddot{u}_r, r$, and ρ are the radial strain, hoop strain, axial strain, shear strain, radial stress, hoop stress, Young modulus of elasticity, Poisson's ratio, thermal expansion, temperature, radial acceleration, radial position, and density, respectively. The root of the inverse coefficient of radial acceleration in Eq. (4) is known as the stress wave velocity,

$$\nu = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(1-2\nu)}} \quad (5)$$

On the other hand, the main governing equations of the thermal energy transfer are.

Fourier heat conduction equation.

Assuming 1-D heat flow (only in the radial direction), the heat conduction through the thickness of the thick-walled cylinder is governed by the 1-D heat conduction equation as [28, 29]:

$$q_r = -kA \frac{\partial T}{\partial r} \quad (6)$$

Second law of thermodynamics.

Considering the energy balance in the absence of the heat source, then it can also be described in one-dimensional as the following [30]:

$$\frac{1}{A} \frac{\partial q_r}{\partial r} = -T_o \frac{dS}{dt} \quad (7)$$

Entropy.

The entropy formula for isotropic material as a function of strains and material properties can be described as [7]:

$$S = \left(\frac{\rho C}{T_o} \right) T + \frac{E\alpha}{(1-2\nu)} \frac{\partial}{\partial t} (\varepsilon_r + \varepsilon_\theta) \quad (8)$$

where q_r, T_o, k, S, C, t , and α^* are, respectively, the heat flux vector in the radial direction, reference temperature, thermal conductivity, entropy, specific heat, time, and thermal diffusivity.

Eliminating $S, q_r, \varepsilon_\theta$, and ε_r using Eqs. (1), (6) (7) and (8), the heat conduction equation considering TDMP can be expressed as:

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{k} \frac{dk}{dr} \frac{\partial T}{\partial r} - \frac{1}{\alpha^*} \frac{\partial T}{\partial t} - \frac{T_o E \alpha}{k(1-2\nu)} \left(\frac{\partial^2 u_r}{\partial t \partial r} + \frac{1}{r} \frac{\partial u_r}{\partial t} \right) = 0 \quad (9)$$

It is worth noting that Eq. (9) is a nonlinear second-order partial differential equation. Also, this term disappears for the case of TIMP; in this case, the equation is classified as a linear second-order partial differential equation.

Equations (4) and (9) together represent the governing equations of the classical coupled thermo-elasticity problem in cylindrical coordinates considering TDMP. Neglecting the strain rate term (last term) in the heat conduction Eq. (9) expresses the governing equations of the dynamic uncoupled thermo-elasticity. In that case, the heat conduction equation is solved independently of displacement Eq. (4).

Boundary conditions (loading conditions) besides initial conditions must be specified to solve Eqs. (4) and (9) simultaneously. On the interior surface of the thick-walled cylinder, the applied thermo-mechanical boundary conditions may be described as [17]:

$$w_{11} u_r(R_i, t) + w_{12} \frac{du_r(R_i, t)}{dr} + w_{13} T(R_i, t) = f_1(t)$$

$$w_{21}T(R_i, t) + w_{22}\frac{dT(R_i, t)}{dr} = f_2(t) \quad (10)$$

while, at the exterior surface, the boundary conditions may be represented as [17],

$$w_{31}u_r(R_o, t) + w_{32}\frac{du_r(R_o, t)}{dr} + w_{33}T(R_o, t) = f_3(t)$$

$$w_{41}T(R_o, t) + w_{42}\frac{dT(R_o, t)}{dr} = f_4(t) \quad (11)$$

The initial conditions of the thick-walled cylinder before applying the thermo-mechanical loads may also be defined as:

$$T(r, 0) = f_5(r)$$

$$u_r(r, 0) = f_6(r) \quad (12)$$

where w_{ij} are the thermal and mechanical loading coefficients describing different boundary conditions such as temperature, flow convection coefficient, heat flux, flow pressure, and displacement.

3 Solution methodology

This study investigates the solution of the classical coupled thermo-elasticity problem considering nonlinear loading conditions and material nonlinearities. As the analytical solution of the nonlinear thermoelasticity problem is a challenging affair, the numerical solution is used to solve Eqs. (3) and (8) that are described in Sect. 2. The numerical solution is based on the FD approach, which converts the continuous equations to a system of linear algebraic equations by discretizing the time and spatial domains. The FD approach is used for its computational power efficiently, and the model is constructed on the two levels of three points Crank-Nicolson scheme. This scheme is unconditionally stable and has a second-order accuracy for both time and spatial domains [23].

3.1 Differential time and space scheme

The solution domain $\bar{D}$ of the Eqs. (3) and (8) is discretized with uniform mesh in spatial and time, described as $\bar{D}_{ij} = \{(r_j, t_n) | j = 1, 2, \dots, J; n = 1, 2, \dots, N : r \in [R_i, R_o], t \in [0, t_{end}]\}$. Assume $\Delta r = (R_o - R_i)/(J - 1)$ as the mesh size in the spatial domain; thus, $r_j = R_i + j\Delta r$, while $\Delta t = t_{end}/(N - 1)$ is the mesh size in the time domain, and thus $t_n = n\Delta t$.

3.2 The Crank-Nicolson finite difference method

Applying Crank-Nicolson finite difference discretization to nonlinear coupled thermo-elasticity Eqs. (3) and (8) with the boundary conditions described in Eqs. (9) and (10) yields:

Equation of motion, Eq. (4):

$$A_j T_{j-1}^{n+1} + B_j T_{j+1}^{n+1} + C_j u_{j-1}^{n+1} + D_j u_j^{n+1} + E_j u_{j+1}^{n+1} \\ = -A_j T_{j-1}^n - B_j T_{j+1}^n - C_j u_{j-1}^n + Q_j u_j^n - E_j u_{j+1}^n + F_j u_{j+1}^{n-1} \quad (13)$$

Heat heat conduction equation, Eq. (9)

$$G_j T_{j-1}^{n+1} + H_j T_j^{n+1} + I_j T_{j+1}^{n+1} + K_j u_{j-1}^{n+1} + L_j u_j^{n+1} + M_j u_{j+1}^{n+1} \\ = -G_j T_{j-1}^n + P_j T_j^n - I_j T_{j+1}^n + K_j u_{j-1}^n + L_j u_j^n + M_j u_{j+1}^n \quad (14)$$

Internal BC, Eq. (10):

$$u_0 = X_1 u_2 + Y_1(t)$$

$$T_0 = X_3(t)T_2 + X_4(t)T_f(t) \quad (15)$$

External BC, Eq. (11):

$$u_{M+1} = X_2 u_{M-1} - Y_2(t)$$

$$T_{m+1} = X_5(t)T_{m-1} + X_6(t)T_\infty \quad (16)$$

The detailed step-by-step derivation has been provided in Appendix A. Equations (13) and (14) may be cast into the matrix form as the following:

$$\begin{bmatrix} [A]_{J \times J} & [B]_{J \times J} \\ [E]_{J \times J} & [F]_{J \times J} \end{bmatrix}_{2J \times 2J} \begin{Bmatrix} \{v_1\}_{J \times 1} \\ \{v_2\}_{J \times 1} \end{Bmatrix}_{2J \times 1}^{n+1} \\ = \begin{bmatrix} [C]_{J \times J} & [D]_{J \times J} \\ [G]_{J \times J} & [H]_{J \times J} \end{bmatrix}_{2J \times 2J} \begin{Bmatrix} \{v_1\}_{J \times 1} \\ \{v_2\}_{J \times 1} \end{Bmatrix}_{2J \times 1}^n + \begin{Bmatrix} \{c_1\}_{J \times 1} \\ \{c_2\}_{J \times 1} \end{Bmatrix} \quad (17)$$

In which matrices A to H are coefficient matrices, vectors $\{v_1\}$, and $\{v_2\}$ are temperature and displacement vectors, and $\{c_1\}$ and $\{c_2\}$ are constant vectors, respectively. The detailed description and components are presented in appendix A. The first and last lines in coefficient matrices (corresponding to the thick-walled cylinder inner and outer boundaries) are determined using the boundary conditions (provided in appendix A).

4 Results and discussions

In this section, the developed numerical model is used to predict thick-walled cylinders' temperature and stress variations due to different loading conditions. In order to

Table 1 Material and geometrical properties

<i>Geometry</i>	
$R_i = 50$ mm	
$R_o = 100$ mm	
<i>Material Properties</i>	
$\rho = 7060$	Kg/m ³
$E = 55.9$	Gpa
$\nu = 0.277$	W/m K
$k = 204$	°C ⁻¹
$\alpha = 10 \times 10^{-6}$	J/Kg K
$C = 903$	
<i>Loading conditions</i>	
$T(r, t) = 200$	°C ⁻¹
$\sigma_r(R_i, t) = 0$	Mpa
$\sigma_r(R_o, t) \approx 0$	Mpa
<i>Initial conditions</i>	
$T(R_i, t) = 21$	°C ⁻¹
$\sigma_r(R_i, t) \approx 0$	Mpa

better present the outputs of the first two examples, the results are presented in non-dimensional forms of time t^* and radius r^* , where $t^* = \frac{vt}{R_o - R_i}$ and $r^* = \frac{r - R_i}{R_o - R_i}$. Consequently, the boundaries of radial position r are changed from $[R_i - R_o]$ to $[0 - 1]$ in the non-dimensional form. On the other hand, the stress wave needs a non-dimensional time of 1 to travel from the inner surface to the outer surface of the cylinder and needs the same time for reflection. In other words, the required time for the upfront stress wave ranges $[0 - 1]$ while it ranges $[1, 2]$ for the reflected wave.

4.1 Thermal load

For the purpose of validating the model results with published studies, consider a thick-walled cylinder under

uniform temperature through its thickness. The cylinder geometry, properties, initial conditions, and loading conditions, as same as published studies [31, 32], are listed in Table 1.

Figure 3 shows the dynamic radial and hoop stresses at the mid-section of the thick-walled cylinder due to thermal load. It is noticed that there is a good agreement between both the proposed model and published studies; therefore, the developed FD model has an acceptable degree of accuracy. In Fig. 3, the first two sudden jumps of stresses at non-dimensional times of 0.5 and 1.5 represents the first upfront stress wave and its reflection, respectively. As from Eq. (4), the stress wave velocity, for this example, is 3.18 km/s. Thus, the upfront stress wave takes a non-dimensional time t^* of 0.5 to reach the mid-radial position. On the other hand, the reflected wave takes a $t^* = 1.5$ to reach the same position.

4.2 Sudden mechanical shock

This example investigates the thermal and elastic behaviors of a thick-walled cylinder made of steel 4130 under internal mechanical shock (sudden internal pressure of 300 MPa). This simple loading condition is selected to clearly investigate the thermal response due to the absence of thermal load. The material and geometrical properties of the cylinder are listed in Table 2, while the initial and boundary conditions are stated in Table 3.

According to the loading conditions stated in Eqs. (9) and (10), the values of w_{ij} are equal to zero except w_{11}, w_{12}, w_{31} , and w_{32} , and their values are: $\frac{1}{R_i} \frac{E\nu}{(1+\nu)(1-2\nu)}$, $\frac{E(1-\nu)}{(1+\nu)(1-2\nu)}$, $\frac{1}{R_o} \frac{E\nu}{(1+\nu)(1-2\nu)}$, and $\frac{E(1-\nu)}{(1+\nu)(1-2\nu)}$, respectively.

Using Eq. (4), the stress wave velocity, for this example, is 6 km/s. Thus, the upfront stress wave takes around $1.03 \mu s$ to propagate through the thickness of the cylinder

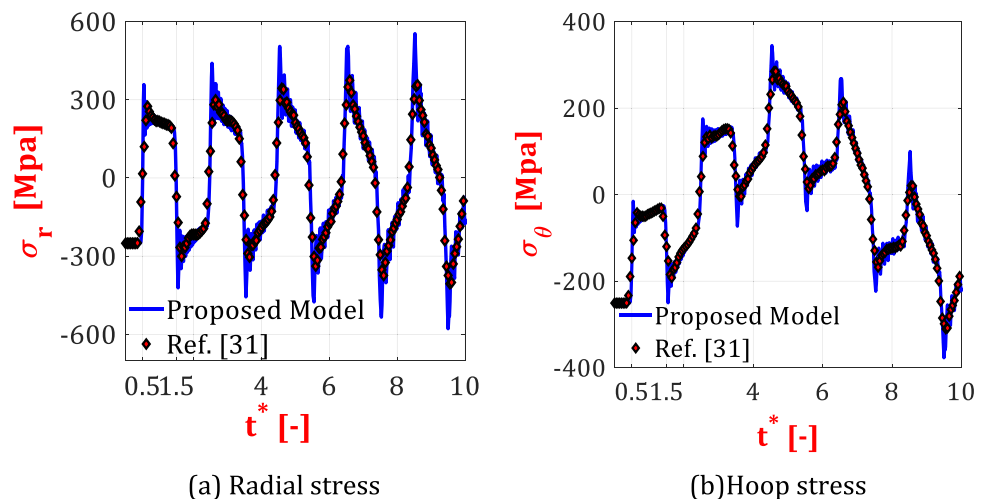
Fig. 3 Stresses-time history at the mid-radial position

Table 2 Material and geometrical properties

<i>Geometry</i>	
$R_i = 3.81 \text{ mm}$	
$R_o = 10 \text{ mm}$	
<i>Material Properties</i>	
$\rho = 7845$	Kg/m^3
$E = 210$	Gpa
$\nu = 0.3$	–
$k = 42.63$	W/m K
$\alpha = 470$	$^{\circ}\text{C}^{-1}$
$C = 10.4 \times 10^{-6}$	J/Kg K

from the inner to the outer surfaces and takes the same period for reflection. Thus, it is better to present the results in the non-dimensional forms of time and radial position.

Figure 4 shows the time-history results for the radial and hoop stresses at non-dimensional radii of 0.25 and 0.75. The radial and hoop stresses at r^* of 0.25 and 0.75 are zero till t^* of 0.25 and 0.75, respectively. As according to the stress wave speed, the stress wave takes t^* of 0.25 and 0.75 to reach r^* of 0.25 and 0.75 from the inner surface, respectively.

Figure 5 depicts the radial stress wave variations through the thickness of the thick-walled cylinder at specified times. The selected non-dimensional times are 0.25, 0.75 for the upfront wave in the solid line and 1.25, 1.75 for the reflected wave in the center line. The upfront stress wave takes nearly t^* of 0.25 and 0.75 to reach non-dimensional radii of 0.25 and 0.75, respectively. On the other hand, the reflected wave takes t^* of 1.75 and 1.25 to reach r^* of 0.25 and 0.75, respectively. From Fig. 5, the t^* between the upfront and the reflected waves to reach r^* of 0.25 and 0.75 equals nearly 1.5 and 0.5. This clearly illustrates the time of action of stresses in Fig. 4 at r^* of 0.25 and 0.75.

It is interesting to note that the quasi-static internal pressure generates compressive radial and tensile hoop stresses throughout the entire thickness of the thick-walled cylinder. However, under the shock pressure load, as shown in Fig. 4(b), the compressive hoop stress starts initially at $t^* = 0.25$, then it approaches the tensile nature after around $t^* = 0.75$.

Figure 6 shows the radial and hoop stress attenuation with more extended time at a non-dimensional radial position of 0.25. Results show that after nearly 2000 stress wave cycles (nearly 2 ms), the radial and hoop stresses reach steady-state conditions with values similar to those under quasi-static pressure (using the hypothesis of static-uncoupled thermo-elasticity). However, before reaching the steady-state conditions, the coupled thermo-elasticity model predicts higher stresses than the static-uncoupled model. Therefore, it is better to consider coupled

Table 3 The boundary and initial conditions

<i>Boundary conditions</i>	
$T(R_i, t) = 21$	$^{\circ}\text{C}^{-1}$
$T(R_o, t) = 21$	$^{\circ}\text{C}^{-1}$
$\sigma_r(R_i, t) = -300$	Mpa
$\sigma_r(R_o, t) \approx 0$	Mpa
<i>Initial conditions</i>	
$T(R_i, t) = 21$	$^{\circ}\text{C}^{-1}$
$\sigma_r(R_i, t) \approx 0$	Mpa

thermo-elasticity models, especially in thick-walled cylinder applications prone to sudden boundary conditions.

Figure 7 depicts the time variation of the temperature waves at non-dimensional radial positions of 0.25 and 0.75. Interestingly, while the thick-walled cylinder is only subjected to a mechanical load, a temperature variation is observed due to internal molecular friction.

4.3 Cyclic time-dependent thermo-mechanical shocks

This example investigates a temperature-dependent thick-walled cylinder due to cyclic time-dependent thermo-mechanical shock (machine gun barrel under continuous shooting). The thick-walled cylinder material is 4130 steel, and its material properties variation with temperature is shown in Fig. 8 [33], while the geometry is listed in the first section of Table 2. Firstly, let's introduce the boundary conditions in some detail, as follows:

4.4 Internal boundary conditions

The inner surface of the thick-walled cylinder is subjected to cyclic nonlinear thermo-mechanical shocks. Each shock consists of a 5 ms loading period, as in Fig. 9, and a 95 ms unloading period to produce a complete shock of 100 ms, as shown in Fig. 10. The time axis in Fig. 10 is put in log scale to clearly depict the first few milliseconds which represent the loading period. These loads are generated due to the burning of 1.63 gm of double-base propellant [36]. As in Fig. 9, the loads' effect reaches after 0.6 ms the position of calculations; this position is far about 178 mm from the load source section (chamber section). Also, this position is the same as the experimental measurement, which will be explained in the next section.

The inner thermal boundary condition is represented, based on Newton's law of cooling and continuity [34], as follows

$$k \left. \frac{\partial T}{\partial r} \right|_{r=R_i} = h_f (T - T_f)$$

Fig. 4 Stresses-time history at specific non-dimensional radial positions

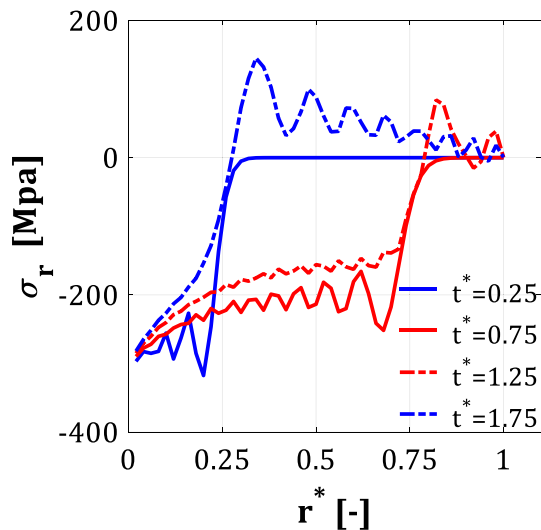
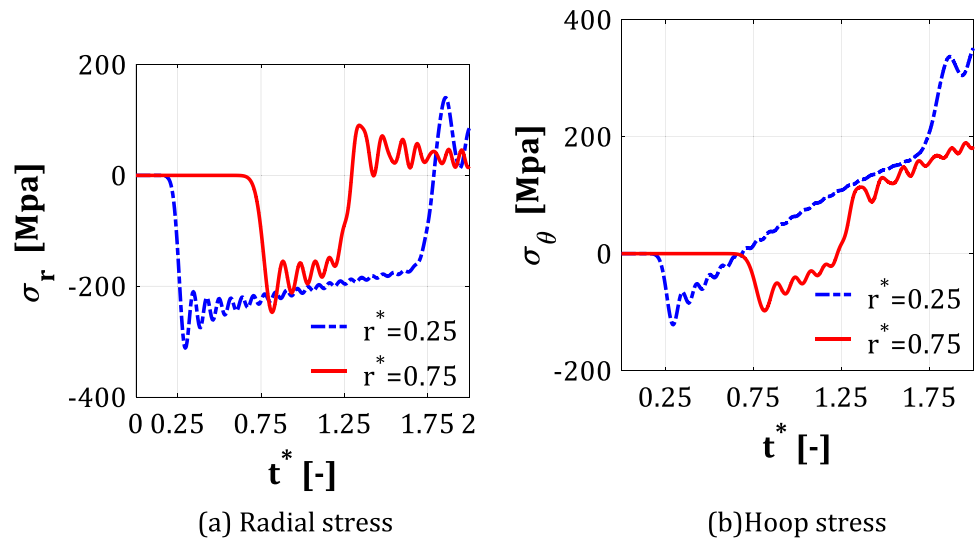


Fig. 5 Radial stress variation through-thickness at specific t^*

while the mechanical boundary condition is represented as:

$$p(R_i, t) = p_f(t)$$

where h_f , T_f , and p_f are the convection coefficient, temperature, and pressure of the inner gas flow.

4.5 External boundary conditions

The outer surface of the thick-walled cylinder is only affected by ambient conditions (no external forces). Consequently, the thermo-mechanical outer boundary condition becomes

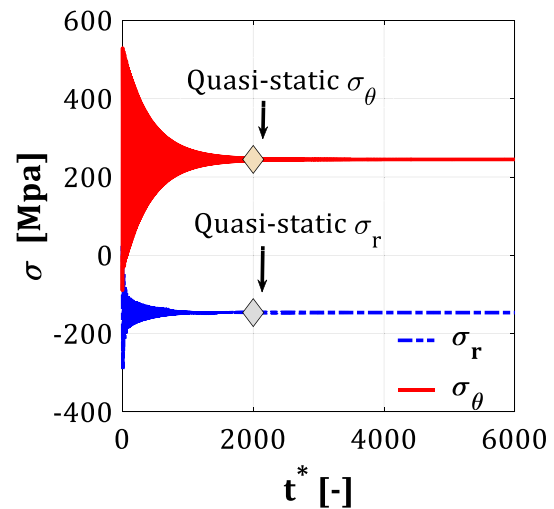


Fig. 6 Radial stress attenuation-time history at specific r^*

$$k \frac{\partial T}{\partial r} \Big|_{r=R_o} = h_\infty (T_\infty - T)$$

$$p(R_o, t) = p_\infty$$

where h_∞ , T_∞ , and p_∞ are the free convection coefficient, ambient temperature, and ambient pressure. The values are considered constant with values of 15 W/m²K, 300 K, and 0.1013 Mpa, respectively.

4.5.1 Thermal accumulation

In fact, the time between two successive input loads is insufficient to allow the cylinder to reach its initial conditions. Consequently, a part of thermal energy is stored

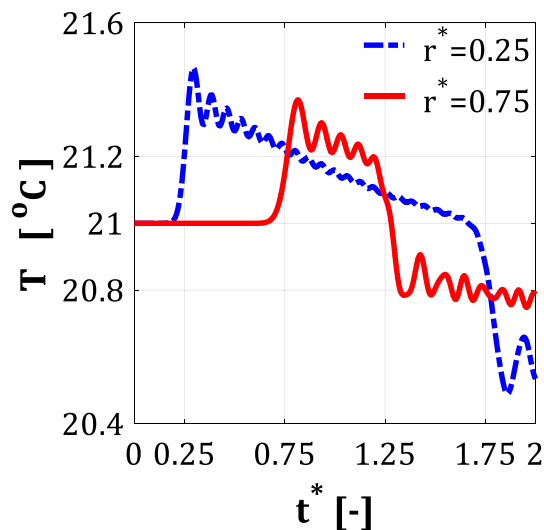


Fig. 7 Temperature–time history at specific non-dimensional radial positions

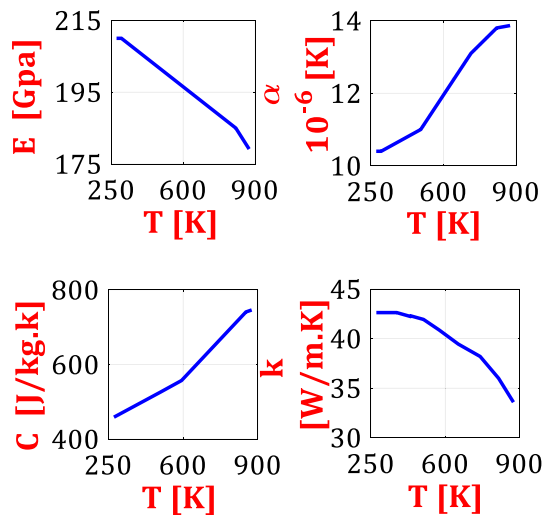
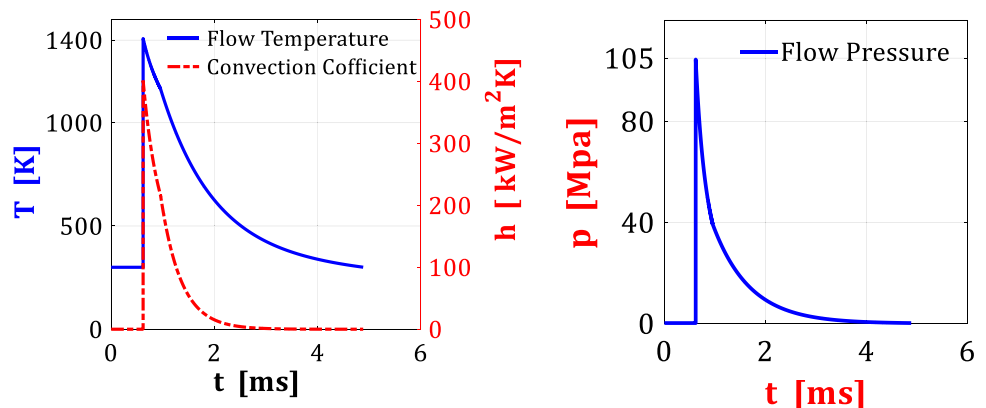


Fig. 8 Material properties of steel 4130 variations with temperature

Fig. 9 Applied thermo-mechanical loads during the loading period at the inner surface of the thick-walled cylinder, 178 mm from the load source section



inside the cylinder thickness, then it accumulates during the successive loading to introduce the so-called “thermal accumulation.” This thermal accumulation causes variations in both stress and temperature distributions through the cylinder thickness. To better realize this, ten successive nonlinear thermo-mechanical shocks are applied at the inner surface of a thick-walled cylinder, each shock as in Fig. 10. The calculations are performed based on the developed model considering the two hypotheses of TIMP and TDMP. The material properties for the case of TIMP are listed in the first section of Table 2, while Fig. 8 represents the material properties for the case of TDMP.

Figure 11 depicts the temperature variation at the thick-walled cylinder’s inner and outer surfaces due to the ten successive time-dependent thermo-mechanical shocks considering both TIMP and TDMP hypotheses. For the two hypotheses, Fig. 12 also shows the temperature variation through the thickness of the thick-walled cylinder after applying the ten successive thermo-mechanical shocks.

As in Fig. 11, the temperature after each cycle does not reach the initial temperature (300 K in this example), which clearly defines the thermal accumulation due to successive thermal cycles. Also, examining the results in Fig. 11 and Fig. 12, the maximum inner surface temperature is 425 and 449 K under the hypothesis of TDMP and TIMP, respectively. Also, the inner surface temperature after applying ten successive cycles is 324.9 and 327 K, while the anticipated temperature at the outer surface is 305 and 307.4 K for TDMP and TIMP, respectively. These values conclude that the temperature values in the case of TDMP are noticeably lower than those in the case of TIMP. Figure 8 illustrates the reason, as the material’s ability to store thermal energy increases with temperature increase while its values of thermal conductivity decrease. In other words, according to the first law of thermodynamics, part of the applied thermal energy will be stored in the cylinder material, and the other part will transfer through the thickness and then to the surrounding. Due to temperature

Fig. 10 Applied thermo-mechanical loads for a complete cycle at the inner surface of the thick-walled cylinder, 178 mm from the load source section

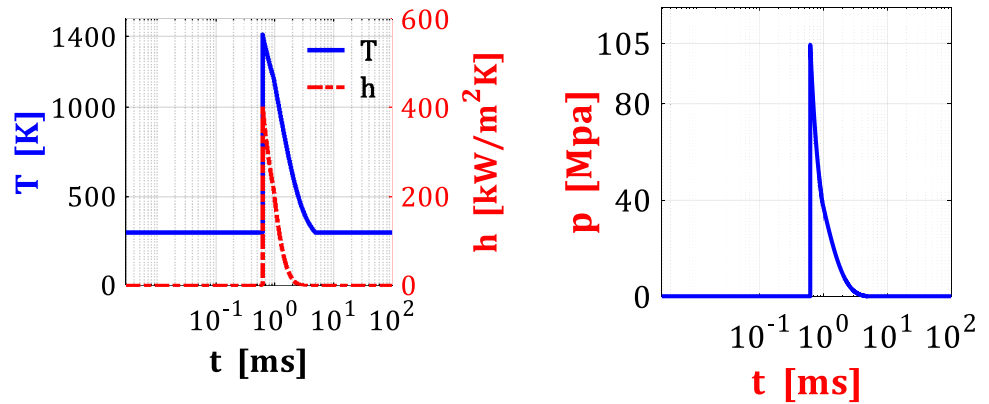


Fig. 11 Temperature–time history of the tube boundaries for 10 cycles

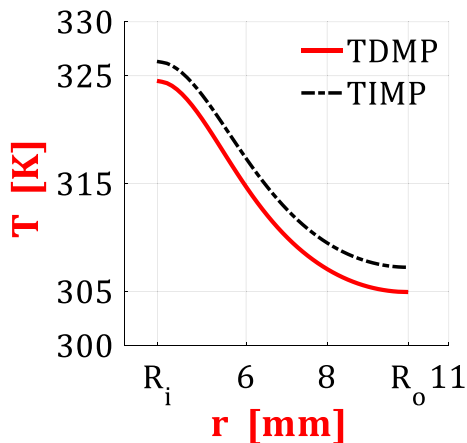
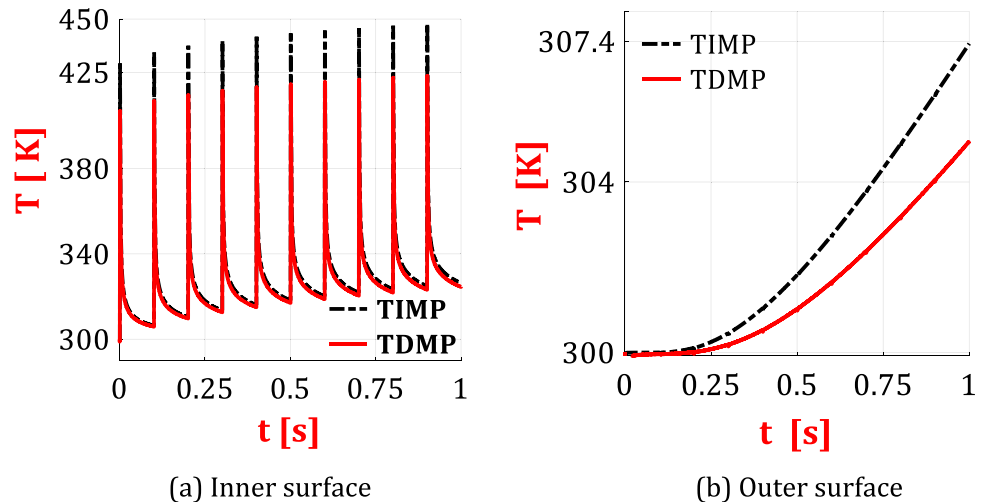


Fig. 12 The temperature variation through the cylinder thickness after the 10 cycles

increases, the cylinder's ability to store heat increases, so the stored energy part increases while the conducted thermal part decreases. Therefore, the temperature values in the case of TDMP are lower than those in the case of TIMP.

Figure 13 depicts the hoop stress variation at the middle-radial surface of the thick-walled cylinder due to ten successive shocks considering both TIMP and TDMP hypotheses. The TIMP hypothesis predicts a maximum compressive radial and tensile hoop stress at the tenth cycle of around 210 and 450 MPa at the mid-wall, while it equals, respectively, 160 and 3370 MPa in the case of TDMP. Therefore the TIMP also overestimates the stress values compared to TDMP. Equation (1) shows the reason, as TIMP overestimates the predicted temperature; therefore, it also overestimates the predicted stresses compared with TDMP.

Adding relaxation periods after successive shocks may be used to control thermal accumulation. To better recognize that, 5 s relaxation time is added after the ten successive loads for the previous loads. Figure 14 displays the temperature variation at a temperature-dependent thick-walled cylinder's inner and outer radii due to ten successive shocks in 6 s. The first second in the time axis represents the time of the ten successive shocks, while the rest of the time represents the 5 s relaxation period. It easily realizes the mitigation of thermal accumulation during

Fig. 13 Stresses-time history at the mid- radial section of the thick-walled cylinder due to 10 cycles

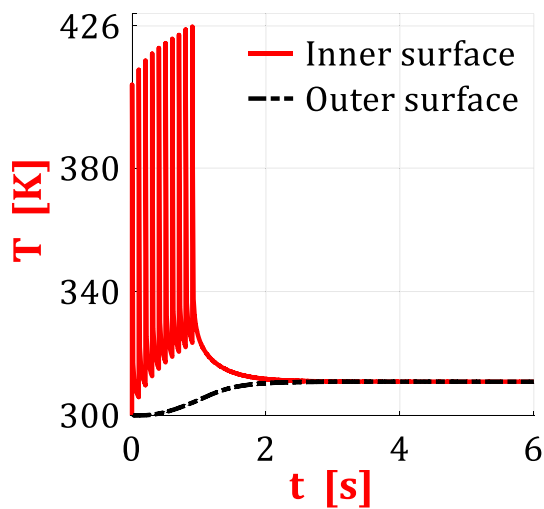
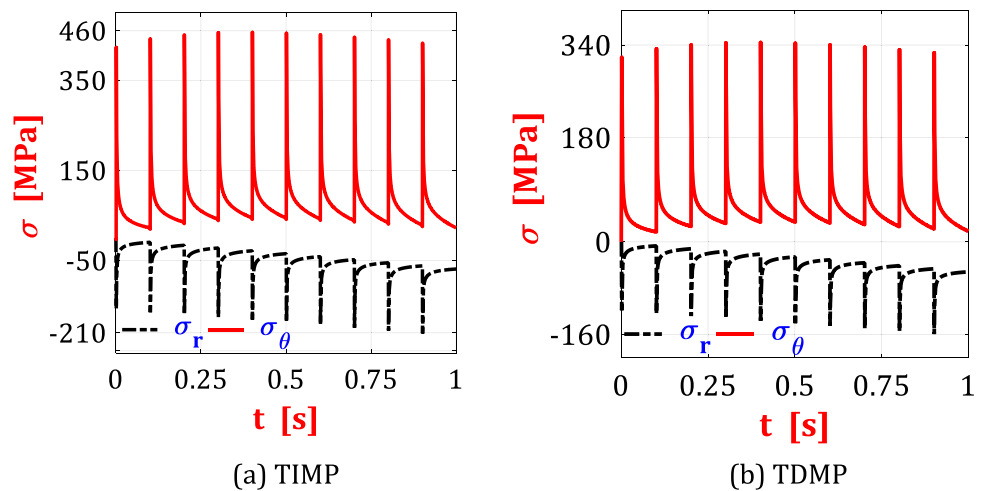


Fig. 14 The temperature variation during the 10 cycles in 6 s under the hypothesis of TDMP

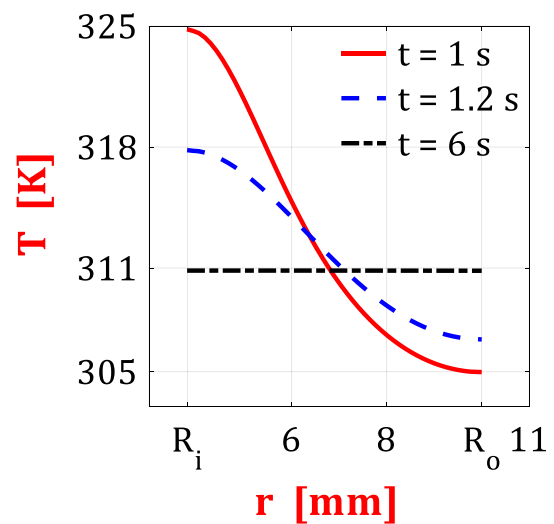


Fig. 15 The temperature profile through the cylinder thickness under the hypothesis of TDMP

the relaxation period. To better illustrate it in more detail, Fig. 15 represents the temperature variation through the cylinder thickness after different relaxation periods. These three plots in Fig. 15 depict the profile at 6 s, 1.2 s, and just after applying ten cycles, corresponding to 5 s, 0.2 s, and 0 s relaxation periods, respectively. The temperature difference between the cylinder surfaces dropped by 7 K by applying a 0.2 s relaxation time, while the temperature is uniformly distributed after a relaxation time of 5 s.

Figure 16 shows the time history of dynamic radial and hoop stresses at the intermediate radius of the thick-walled cylinder due to the prementioned loads. Although, after the relaxation period, the temperature is uniformly distributed, as in Fig. 15, the stresses did not vanish. This is because the temperature of the thick-walled cylinder is not reached the initial temperature. Also, it is interesting

to mention that the residual temperature generates compressive residual hoop stress, which enhances the thick-walled cylinder's performance [35]. Therefore, from the thick-walled cylinder design point of view, the TDMP hypothesis is better than the TIMP hypothesis as it underestimates compressive residual hoop stress variation through the cylinder thickness. Also, the TDMP hypothesis is better than the TIMP hypothesis if it can predict accurate results compared with the TIMP hypothesis.

4.5.2 Experimental study

An experimental study has been designed to further investigate the impact of considering the TDMP hypothesis on the accuracy of results. The measurements are performed on a real machine gun barrel under burst shooting. The

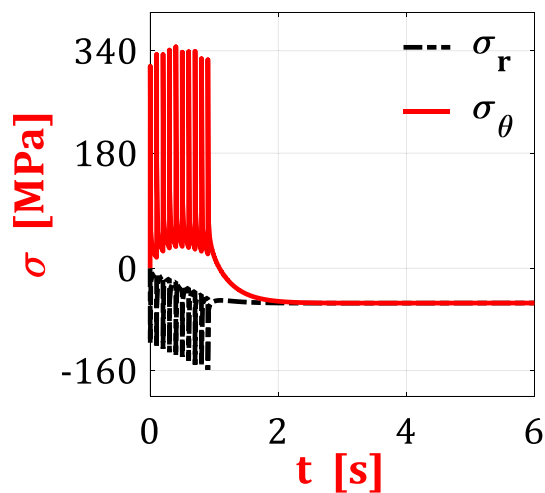


Fig. 16 Stresses-time history at the mid- radial section of the temperature-dependent thick-walled cylinder due to 10 cycles in 6 s

barrel has the same dimensions and material properties as those in Table 2 and Fig. 8. The sketch of the experimental setup is shown in Fig. 17.

The inner surface of the cylinder is subjected to 85 cyclic nonlinear thermo-mechanical shocks with a particular regime of fire. The regime of fire consists of 3 bursts of shocks, two bursts of 30, and a burst of 25. Each burst is followed by a relaxation period, as shown in Fig. 18.

Each thermo-mechanical shock represents the loads due to the propellant burning, as shown in Fig. 10 [36]. The outer surface temperature of the cylinder is measured using a highly sensitive thermocouple type K (Chrome-Aluminium) with 0.5 mm. It is too difficult to directly fix the thermocouple at the axial section of the load source (chamber section); consequently, the measurement is

performed at a distance of 178 mm from the load source section. The thermocouple signal is processed using a data acquisition system with a maximum sampling rate of 200 kHz. The relaxation time after each burst of shocks is the time consumed during the reloading of new bolts. Also, this relaxation time is necessary to enhance the measurement results as the duration of the applied loads is too short.

Figure 19 shows the predicted temperature-time history at the inner and outer surfaces of the temperature-dependent barrel cylinder due to 85 nonlinear thermo-mechanical shocks. The transient temperature spikes for each successive burst of thermo-mechanical shocks appear as one thick line for each burst due to the used time scale. The temperature peak always increases between each subsequent shock due to heat accumulation. During the relaxation period, the temperature difference between the inner and outer surfaces of the thick-walled cylinder becomes negligible.

Figure 20 depicts the time history of the dynamic radial and hoop stresses at the intermediate surface of the thick-walled cylinder. It is noteworthy to point out that, as in Fig. 20(b), the hoop stress with time pass becomes compressive due to the effect of thermal loads. On the other hand, the stress spikes almost have the same length of around 340 MPa, and it is always upward. In fact, this is the effect of mechanical loads. In brief, the mechanical load effect is not changed with subsequent cycles, and it has a maximum tensile hoop and compressive radial stresses of 315 and 105 MPa, respectively. In comparison, the thermal load causes the shifting down of these values and changes the hoop stress from tensile to compressive.

Figure 21 shows a thick-walled cylinder's measured and predicted outer surface temperature-time history due

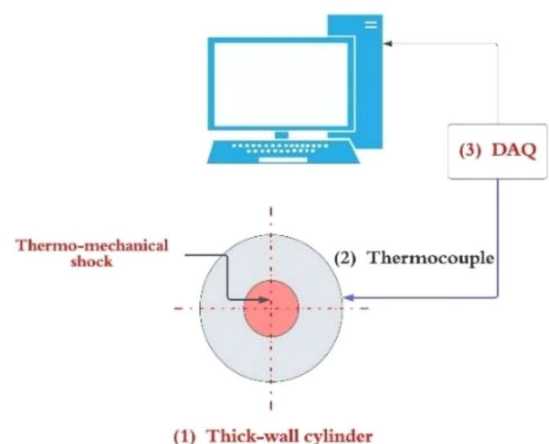
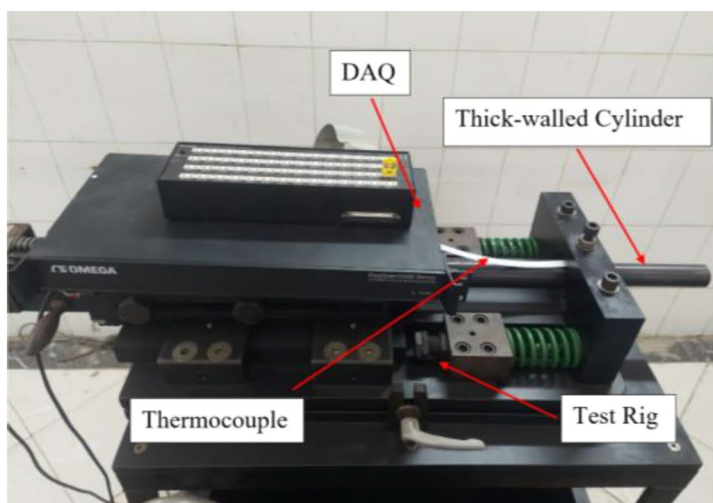


Fig. 17 Schematic drawing of the experimental setup

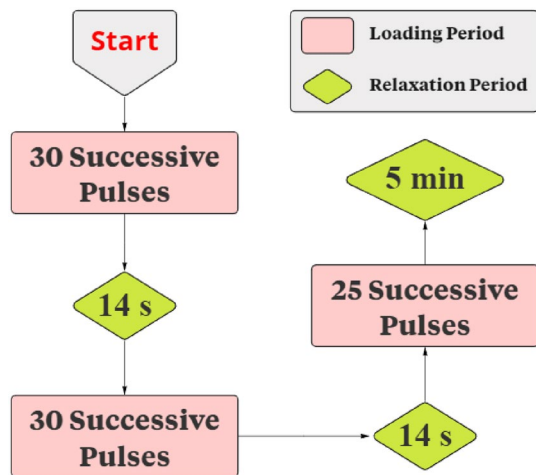


Fig. 18 Schematic representation of applied thermo-mechanical shocks

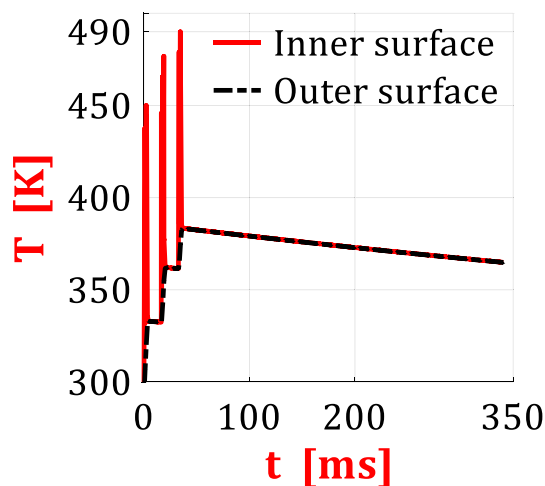
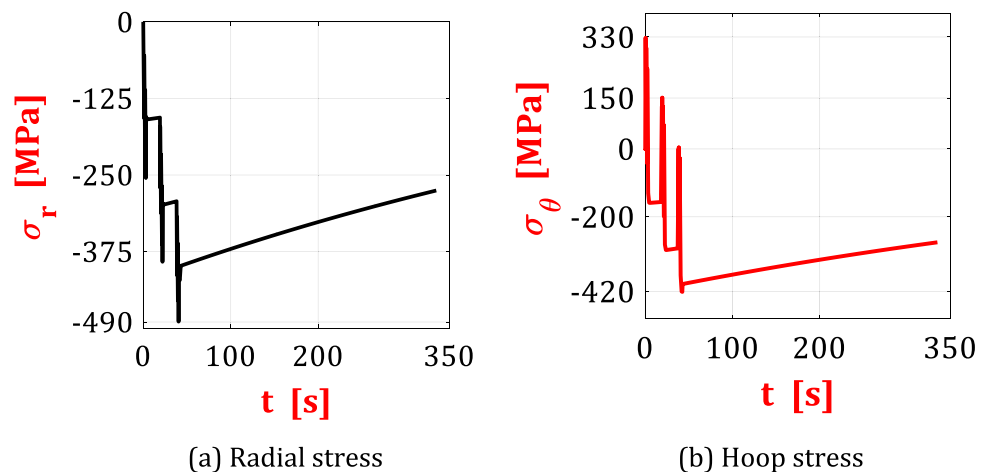


Fig. 19 Inner and outer surface temperatures for applying 85 shocks under the hypothesis of TDMP

Fig. 20 Stresses-time history at the mid-radial section of the temperature-dependent barrel cylinder due to 85 thermo-mechanical shock pulses



to the predefined regime. The predicted temperature is performed considering the two hypotheses of TDMP and TIMP. The results show a good agreement between calculated results based on TDMP and experimental results. Simultaneously, model-based TIMP could not accurately capture the thermal response, particularly at the end of the third burst of shocks and after that. The maximum measured temperature is 380 K, compared with 385 and 404 K for TDMP and TIMP, respectively. In comparison, the measured temperature after the completion of the load cycles and subsequent relaxation is 358 K, compared with 364 and 374 K for TDMP and TIMP, respectively. The maximum error between model-based TDMP and experiment was less than 3%, while that between the model-based on TIMP and experiment was around 10%.

Figure 22 displays the dynamic hoop stress's time history at the thick-walled cylinder's intermediate surface, considering TIMP. As in Fig. 22 and Fig. 20(b), the maximum compressive hoop stress is 520 and 420 MPa for TIMP and TDMP, respectively. In fact, Figs. 20(b) 21 and 22 highlight the positive impact of considering TDMP on the accuracy of results which also guarantees an accurate thick-walled cylinder design.

5 Conclusion

The main subject of this study is to investigate the coupled thermo-elastic behaviors in temperature-dependent thick-walled cylinders under cyclic nonlinear thermo-mechanical loads. To achieve this, firstly, the governing equations addressing the classical coupled thermo-elasticity in cylindrical coordinates considering material nonlinearities are formulated. Then a detailed numerical solution is presented to predict the displacement and temperature fields. The proposed numerical solution is based on the implicit finite difference using the Crank-Nicolson scheme.

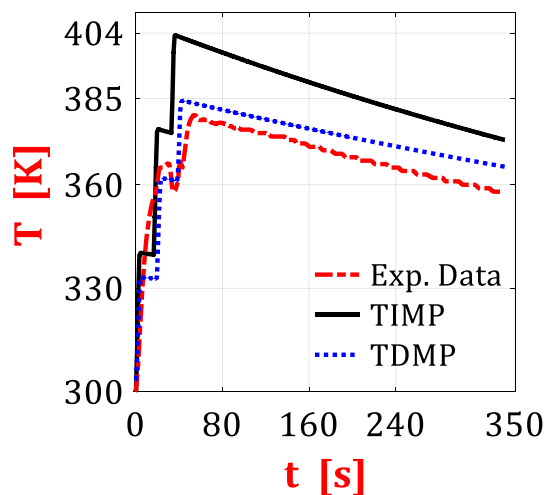


Fig. 21 Temperature–time history of the outer surface of the temperature-dependent thick-walled cylinder due to 85 shocks

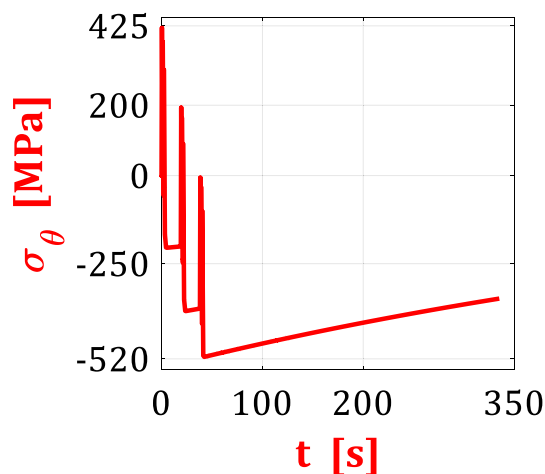


Fig. 22 Hoop stress–time history at the mid- radial section of the thick-walled cylinder due to 85 pulses under the hypothesis of TIMP

This proposed solution of the classical coupled thermo-elasticity represents the temperature and stress fields to a reasonable level of accuracy for different boundary conditions. The model reveals that, before reaching the steady state condition, the coupled thermo-elasticity model predicts higher thermo-elastic behaviors than the uncoupled models. Thus, it is better to consider the coupled model in structures under sudden boundary conditions.

Secondly, an experimental setup is designed to deeply compare the TDMP model results and the TIMP model results with experimental data. Therefore, the outer surface temperature of a machine gun barrel is measured using a highly sensitive thermocouple. The machine gun is under burst shooting. Based on the temperature measurement,

the model results had good agreement with the experimental data in the case of TDMP, with a maximum error of 3% compared with a maximum error of 10% for the case of TIMP. Also, TIMP overestimates temperature and stresses, which badly affects thick-walled cylinder design.

Finally, to the best of the authors' knowledge, this is the first time that the classical nonlinear coupled thermo-elasticity has been investigated under cyclic nonlinear thermo-mechanical loads. Moreover, the comparative study has been conducted between two hypotheses, TIMP and TDMP. More generally, the proposed model can be used for trade studies to design lightweight thick-walled cylinders according to their severe operating conditions.

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Ethical approval This article does not contain any studies with human participants performed by any of the authors

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Appendix

The Crank-Nicolson finite difference discretization for Eqs. (4) and (9) with mixed boundary conditions (10) and (11) is given by:

For the radial displacement $u(r, t)$ Eq. (4)

$$\frac{\partial^2 u_r}{\partial r^2} + \frac{1}{r} \frac{\partial u_r}{\partial r} - \frac{u_r}{r^2} - \frac{\alpha(1+\nu)}{(1-\nu)} \frac{\partial T}{\partial r} - \frac{\rho(1+\nu)(1-2\nu)}{E(1-\nu)} \ddot{u}_r = 0$$

Applying the Crank-Nicolson finite difference discretization,

$$\begin{aligned} & \frac{1}{2\Delta r^2} (u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1} + u_{j+1}^n - 2u_j^n + u_{j-1}^n) \\ & + \frac{1}{4\Delta r} \frac{1}{r_j} (u_{j+1}^{n+1} - u_{j-1}^{n+1} + u_{j+1}^n - u_{j-1}^n) \\ & - \frac{1}{4r_j^2} (u_{j+1}^{n+1} + u_{j-1}^{n+1} + u_{j+1}^n + u_{j-1}^n) - \frac{\alpha(1+\nu)}{(1-\nu)} \frac{1}{4\Delta r} (T_{j+1}^{n+1} - T_{j-1}^{n+1} + T_{j+1}^n - T_{j-1}^n) \\ & - \frac{\rho(1+\nu)(1-2\nu)}{E(1-\nu)} \frac{1}{\Delta t^2} (u_j^{n+1} - 2u_j^n + u_j^{n-1}) = 0 \end{aligned}$$

After rearranging next time step ($n+1$) in one side and the previous time steps in the other sides the displacement equation will be,

$$\begin{aligned} & \Rightarrow \underbrace{\frac{\alpha(1+\nu)}{(1-\nu)} \frac{1}{4\Delta r} T_{j-1}^{n+1}}_{A_j} - \underbrace{\frac{\alpha(1+\nu)}{(1-\nu)} \frac{1}{4\Delta r} T_{j+1}^{n+1}}_{B_j} \\ & + \underbrace{\left(\frac{1}{2\Delta r^2} - \frac{1}{4\Delta r} \frac{1}{r_j} - \frac{1}{4r_j^2} \right) u_{j-1}^{n+1}}_{C_j} - \underbrace{\left(\frac{1}{r^2} + \frac{\rho(1+\nu)(1-2\nu)}{E(1-\nu)} \frac{1}{\Delta t^2} \right) u_j^{n+1}}_{D_j} \\ & + \underbrace{\left(\frac{1}{2\Delta r^2} + \frac{1}{4\Delta r} \frac{1}{r_j} - \frac{1}{4r_j^2} \right) u_{j+1}^{n+1}}_{E_j} \\ & = - \underbrace{\frac{\alpha(1+\nu)}{(1-\nu)} \frac{1}{4\Delta r} T_{j-1}^n}_{-A_j} + \underbrace{\frac{\alpha(1+\nu)}{(1-\nu)} \frac{1}{4\Delta r} T_{j+1}^n}_{-B_j} - \underbrace{\left(\frac{1}{2\Delta r^2} - \frac{1}{4\Delta r} \frac{1}{r_j} - \frac{1}{4r_j^2} \right) u_{j-1}^n}_{-C_j} \\ & + \underbrace{\left(\frac{1}{\Delta r^2} - \frac{\rho(1+\nu)(1-2\nu)}{E(1-\nu)} \frac{2}{\Delta t^2} \right) u_j^n}_{Q_j} - \underbrace{\left(\frac{1}{2\Delta r^2} + \frac{1}{4\Delta r} \frac{1}{r_j} - \frac{1}{4r_j^2} \right) u_{j+1}^n}_{-E_j} \\ & + \underbrace{\frac{\rho(1+\nu)(1-2\nu)}{E(1-\nu)} \frac{1}{\Delta t^2} u_j^{n-1}}_{F_j} \\ & A_j T_{j-1}^{n+1} + B_j T_{j+1}^{n+1} + C_j u_{j-1}^{n+1} \\ & + D_j u_j^{n+1} + E_j u_{j+1}^{n+1} = -A_j T_{j-1}^n \\ & - B_j T_{j+1}^n - C_j u_{j-1}^n + Q_j u_j^n - E_j u_{j+1}^n + F_j u_j^{n-1} \end{aligned} \quad (A.1)$$

For the radial temperature equation $T(r,t)$ Eq. (9)

$$\frac{\partial^2 T}{\partial r^2} + \left(\frac{1}{r} + \frac{1}{k} \frac{dk}{dr} \right) \frac{\partial T}{\partial r} - \frac{1}{\alpha^*} \frac{\partial T}{\partial t} - \frac{T_o E \alpha}{k(1-2\nu)} \left(\frac{\partial^2 u_r}{\partial t \partial r} + \frac{1}{r} \frac{\partial u_r}{\partial t} \right) = 0$$

Applying the Crank-Nicolson finite difference discretization,

$$\begin{aligned} & \Rightarrow \frac{1}{2\Delta r^2} (T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1} + T_{j+1}^n - 2T_j^n + T_{j-1}^n) \\ & + \left(\frac{1}{r_j} + \frac{1}{k_j} \frac{k_{j+1} - k_{j-1}}{2\Delta r} \right) \frac{1}{4\Delta r} (T_{j+1}^{n+1} - T_{j-1}^{n+1} + T_{j+1}^n - T_{j-1}^n) \\ & - \frac{1}{\alpha^*} \frac{1}{\Delta t} (T_j^{n+1} - T_j^n) \\ & - \frac{T_o E \alpha}{k(1-2\nu)} \frac{1}{\Delta t} \left[\frac{1}{2\Delta r} (u_{j+1}^{n+1} - u_{j-1}^{n+1} - u_{j+1}^n + u_{j-1}^n) \right. \\ & \left. + \frac{1}{r_j} (u_j^{n+1} - u_j^n) \right] = 0 \end{aligned}$$

After rearranging next time step ($n+1$) in one side and the previous time steps in the other sides the heat equation will be,

$$\begin{aligned} & \underbrace{\left(\frac{1}{2\Delta r^2} - Z_j \frac{1}{4\Delta r} \right) T_{j-1}^{n+1}}_{G_j} - \underbrace{\left(\frac{1}{\Delta r^2} + \frac{1}{\alpha^*} \frac{1}{\Delta t} \right) T_j^{n+1}}_{H_j} \\ & + \underbrace{\left(\frac{1}{2\Delta r^2} + Z_j \frac{1}{4\Delta r} \right) T_{j+1}^{n+1}}_{I_j} \\ & + \underbrace{\left(\frac{T_o E \alpha}{k(1-2\nu)} \frac{1}{2\Delta r \Delta t} \right) u_{j-1}^{n+1}}_{K_j} - \underbrace{\left(\frac{T_o E \alpha}{k(1-2\nu)} \frac{1}{\Delta t} \frac{1}{r_j} \right) u_j^{n+1}}_{L_j} \\ & - \underbrace{\left(\frac{T_o E \alpha}{k(1-2\nu)} \frac{1}{2\Delta r \Delta t} \right) u_{j+1}^{n+1}}_{M_j} = - \underbrace{\left(\frac{1}{2\Delta r^2} - Z_j \frac{1}{4\Delta r} \right) T_{j-1}^n}_{-G_j} \\ & + \underbrace{\left(\frac{1}{\Delta r^2} - \frac{1}{\alpha^* \Delta t} \right) T_j^n}_{P_j} - \underbrace{\left(\frac{1}{2\Delta r^2} + Z_j \frac{1}{4\Delta r} \right) T_{j+1}^n}_{-I_j} \\ & + \underbrace{\left(\frac{T_o E \alpha}{k(1-2\nu)} \frac{1}{2\Delta r \Delta t} \right) u_{j-1}^n}_{K_j} - \underbrace{\left(\frac{T_o E \alpha}{k(1-2\nu)} \frac{1}{\Delta t} \frac{1}{r_j} \right) u_j^n}_{L_j} \\ & - \underbrace{\left(\frac{T_o E \alpha}{k(1-2\nu)} \frac{1}{2\Delta r \Delta t} \right) u_{j+1}^n}_{M_j} \end{aligned}$$

$$\begin{aligned} & G_j T_{j-1}^{n+1} + H_j T_j^{n+1} + I_j T_{j+1}^{n+1} + K_j u_{j-1}^{n+1} + L_j u_j^{n+1} + M_j u_{j+1}^{n+1} \\ & = -G_j T_{j-1}^n + P_j T_j^n - I_j T_{j+1}^n + K_j u_{j-1}^n + L_j u_j^n + M_j u_{j+1}^n \end{aligned} \quad (A.2)$$

$$\text{where, } Z_j = \frac{1}{k_j} \frac{k_{j+1} - k_{j-1}}{2\Delta r}.$$

Internal boundary conditions Eq. (10)**Thermal boundary conditions**

$$w_{21} \frac{T_2 + T_0}{2} + w_{22} \frac{T_2 - T_0}{2\Delta r} = f_2(t)$$

$$T_0 = Y_1 - Y_2 T_2 \quad (\text{A.3})$$

Mechanical boundary conditions

$$w_{11} \frac{u_2 + u_0}{2R_i} + w_{12} \frac{u_2 - u_0}{2\Delta r} + w_{13} \frac{T_2 + T_0}{2} = f_1(t)$$

Using the expression of T_0 from Eq. (A.3)

$$u_0 = X_1 + X_2 T_2 - X_3 u_2 \quad (\text{A.4})$$

$$X_1 = (2f_1(t) - w_{13}Y_1) / \left(\frac{w_{11}}{R_i} - \frac{w_{12}}{\Delta r} \right),$$

$$X_2 = w_{13}(Y_2 - 1) / \left(\frac{w_{11}}{R_i} - \frac{w_{12}}{\Delta r} \right),$$

$$X_3 = \left(\frac{w_{11}}{R_i} + \frac{w_{12}}{\Delta r} \right) / \left(\frac{w_{11}}{R_i} - \frac{w_{12}}{\Delta r} \right),$$

$$\text{where, } Y_1 = 2f_2(t) / \left(w_{21} - \frac{w_{22}}{\Delta r} \right),$$

$$Y_2 = \left(w_{21} + \frac{w_{22}}{\Delta r} \right) / \left(w_{21} - \frac{w_{22}}{\Delta r} \right).$$

External boundary conditions Eq. (11)**Thermal boundary conditions**

$$w_{41} \frac{T_{M+1} + T_{M-1}}{2} + w_{42} \frac{T_{M+1} - T_{M-1}}{2\Delta r} = f_4(t)$$

$$T_{M+1} = Y_3 - Y_4 T_{M-1} \quad (\text{A.5})$$

Mechanical boundary conditions

$$w_{31} \frac{u_{M+1} + u_{M-1}}{2R_o} + w_{32} \frac{u_{M+1} - u_{M-1}}{2\Delta r} + w_{33} \frac{T_{M+1} + T_{M-1}}{2} = f_3(t)$$

Using the expression of T_{M+1} from Eq. (A.5)

$$u_{M+1} = X_4 + X_5 T_{M-1} - X_6 u_{M-1} \quad (\text{A.6})$$

$$X_5 = w_{33}(Y_4 - 1) / \left(\frac{w_{31}}{R_o} + \frac{w_{32}}{\Delta r} \right),$$

$$X_6 = \left(\frac{w_{31}}{R_o} - \frac{w_{32}}{\Delta r} \right) / \left(\frac{w_{31}}{R_o} + \frac{w_{32}}{\Delta r} \right),$$

$$Y_3 = 2f_4(t) / \left(w_{41} + \frac{w_{42}}{\Delta r} \right),$$

$$Y_4 = \left(w_{41} - \frac{w_{42}}{\Delta r} \right) / \left(w_{41} + \frac{w_{42}}{\Delta r} \right), \text{ and}$$

where,

$$X_4 = (2f_3(t) - w_{33}Y_3) / \left(\frac{w_{31}}{R_o} + \frac{w_{32}}{\Delta r} \right).$$

The boundary conditions are applied for left side (next time step) and right side (current step) for the displacement and radial Eqs. (A.1) and (A.2), as follows

At the thick-walled cylinder inner surface R_i ($j = 1$).**For the radial displacement,**

$$A_1 T_0^{n+1} + B_1 T_2^{n+1} + C_1 u_0^{n+1} + D_1 u_1^{n+1} + E_1 u_2^{n+1} \\ = -A_1 T_0^n - B_1 T_2^n - C_1 u_0^n + Q_1 u_1^n - E_1 u_2^n + F_1 u_1^{n-1}$$

Substituting by the expressions of T_0 and u_0 using Eqs (A.3) and (A.4) into the radial displacement equation and heat equation at the inner surface of the thick-walled cylinder can be written as,

$$\overbrace{(B_1 + C_1 X_2 - A_1 Y_2)}^{b_1} T_2^{n+1} + \overbrace{D_1}^{d_1} u_1^{n+1} + \overbrace{(E_1 - C_1 X_3)}^{e_1} u_2^{n+1} \\ = -\overbrace{(B_1 + C_1 X_2 - A_1 Y_2)}^{b_1} T_2^n \\ + \overbrace{Q_1}^{q_1} u_1^n - \overbrace{(E_1 - C_1 X_3)}^{e_1} u_2^n \\ + \overbrace{F_1 u_1^{n-1} - C_1 (X_1^{n+1} + X_1^n) - A_1 (Y_1^{n+1} + Y_1^n)}^{f_1}$$

$$b_1 T_2^{n+1} + d_1 u_1^{n+1} + e_1 u_2^{n+1} = -b_1 T_2^n + q_1 u_1^n - e_1 u_2^n + f_1 \quad (\text{A.7})$$

Similarly for heat equation,

$$\begin{aligned}
 &\Rightarrow \overbrace{H_1}^{h_1} T_1^{n+1} + \overbrace{(I_1 - G_1 Y_2 + K_1 X_2)}^{i_1} T_2^{n+1} \\
 &+ \overbrace{L_1}^{l_1} u_1^{n+1} + \overbrace{(M_1 - K_1 X_3)}^{m_1} u_2^{n+1} \\
 &= \overbrace{P_1}^{p_1} T_1^n + \overbrace{(G_1 Y_2 - I_1 + K_1 X_2)}^{v_1} T_2^n \\
 &+ L_1 u_1^n + \overbrace{(M_1 - K_1 X_3) u_2^n - G_1 (Y_1^{n+1} + Y_1^n) - K_1 (X_1^{n+1} - X_1^n)}^{s_1} \\
 &h_1 T_1^{n+1} + i_1 T_2^{n+1} + l_1 u_1^{n+1} + m_1 u_2^{n+1} \\
 &= p_1 T_1^n + v_1 T_2^n + l_1 u_1^n + m_1 u_2^n + s_1
 \end{aligned} \quad (A.8)$$

At the thick-walled cylinder outer surface R_j ($j=J$):

For the radial displacement,

$$\begin{aligned}
 &A_J T_{J-1}^{n+1} + B_J T_{J+1}^{n+1} + C_J u_{J-1}^{n+1} + D_J u_{J+1}^{n+1} + E_J u_{J+1}^{n+1} \\
 &= -A_J T_{J-1}^n - B_J T_{J+1}^n - C_J u_{J-1}^n + Q_J u_J^n - E_J u_{J+1}^n + F_J u_J^{n-1}
 \end{aligned}$$

Substituting by the expressions of T_{J+1} and u_{J+1} using Eqs (A.5) and (A.6) into the radial displacement equation at the outer surface of the thick-walled cylinder can be written as,

$$\begin{aligned}
 &\overbrace{(A_J - Y_4 B_J + X_5 E_J)}^{a_J} T_{J-1}^{n+1} + \overbrace{(C_J - X_6 E_J)}^{c_J} u_{J-1}^{n+1} + \overbrace{D_J}^{d_J} u_J^{n+1} \\
 &= -\overbrace{(A_J - Y_4 B_J + X_5 E_J)}^{a_J} T_{J-1}^n - \overbrace{(C_J - X_6 E_J)}^{c_J} u_{J-1}^n + \overbrace{Q_J}^{q_J} u_J^n \\
 &+ \overbrace{F_J u_J^{n-1} - E_J (X_4^{n+1} + X_4^n) - B_J (Y_3^{n+1} + Y_3^n)}^{f_J}
 \end{aligned}$$

$$a_J T_{J-1}^{n+1} + c_J u_{J-1}^{n+1} + d_J u_J^{n+1} = -a_J T_{J-1}^n - c_J u_{J-1}^n + q_J u_J^n + f_J \quad (A.9)$$

Similarly for heat equation,

$$\begin{aligned}
 &\overbrace{(G_J - Y_4 I_J + X_5 M_J)}^{g_J} T_{J-1}^{n+1} + \overbrace{H_J}^{h_J} T_J^{n+1} \\
 &+ \overbrace{(K_J - X_6 M_J)}^{k_J} u_{J-1}^{n+1} + \overbrace{L_J}^{l_J} u_J^{n+1} \\
 &= \overbrace{(Y_4 I_J - G_J + X_5 M_J)}^{v_J} T_{J-1}^n + \overbrace{P_J}^{p_J} T_J^n + \overbrace{(K_J - X_6 M_J) u_{J-1}^n}^{s_J} \\
 &+ L_J u_J^n - I_J (Y_3^{n+1} + Y_3^n) - M_J (X_4^{n+1} - X_4^n)
 \end{aligned}$$

$$\begin{aligned}
 &g_J T_{J-1}^{n+1} + h_J T_J^{n+1} + k_J u_{J-1}^{n+1} + l_J u_J^{n+1} \\
 &= v_J T_{J-1}^n + p_J T_J^n + k_J u_{J-1}^n + l_J u_J^n + s_J
 \end{aligned} \quad (A.10)$$

The radial displacement and heat equations can be written in a very convenient structure written in a matrix form, as:

$$\begin{aligned}
 &\begin{bmatrix} [A]_{J \times J} & [B]_{J \times J} \\ [E]_{J \times J} & [F]_{J \times J} \end{bmatrix}_{2J \times 2J} \begin{Bmatrix} \{v_1\}_{J \times 1} \\ \{v_2\}_{J \times 1} \end{Bmatrix}_{2J \times 1}^{n+1} \\
 &= \begin{bmatrix} [C]_{J \times J} & [D]_{J \times J} \\ [G]_{J \times J} & [H]_{J \times J} \end{bmatrix}_{2J \times 2J} \begin{Bmatrix} \{v_1\}_{J \times 1} \\ \{v_2\}_{J \times 1} \end{Bmatrix}_{2J \times 1}^n \\
 &+ \begin{Bmatrix} \{c_1\}_{J \times 1} \\ \{c_2\}_{J \times 1} \end{Bmatrix}
 \end{aligned}$$

where $[X]$, $X \in \{A-H\}$, $\{v_1\}$, $\{v_2\}$, $\{c_1\}$, $\{c_2\}$ values are as follows:

$$\begin{aligned}
 & \left[\begin{array}{c} \text{Eq. 4} \cdot \left\{ \begin{array}{l} R_i \rightarrow \\ R_o \rightarrow \end{array} \right. \left[\begin{array}{c|c} \overbrace{\begin{matrix} 0 & b_1 & & \\ A_2 & 0 & B_2 & \\ \vdots & \vdots & \vdots & \\ A_{J-1} & 0 & B_{J-1} & \\ & a_J & 0 & \end{matrix}}^{[A]_{J \times J}} & \overbrace{\begin{matrix} d_1 & e_1 & & \\ C_2 & D_2 & E_2 & \\ \vdots & \vdots & \vdots & \\ C_{J-1} & D_{J-1} & E_{J-1} & \\ & c_J & d_J & \end{matrix}}^{[B]_{J \times J}} \\ \hline \overbrace{\begin{matrix} h_1 & i_1 & & \\ G_2 & H_2 & I_2 & \\ \vdots & \vdots & \vdots & \\ G_{J-1} & H_{J-1} & I_{J-1} & \\ & g_J & h_J & \end{matrix}}^{[E]_{J \times J}} & \overbrace{\begin{matrix} l_1 & m_1 & & \\ K_2 & L_2 & M_2 & \\ \vdots & \vdots & \vdots & \\ K_{J-1} & L_{J-1} & M_{J-1} & \\ & k_J & l_J & \end{matrix}}^{[F]_{J \times J}} \end{array} \right]_{J \times 2J} \left[\begin{array}{c} \left\{ \begin{array}{c} v_1 \\ T_1 \\ T_2 \\ \vdots \\ T_{J-1} \\ T_J \end{array} \right\}_{J \times 1} \\ \left\{ \begin{array}{c} v_2 \\ u_1 \\ u_2 \\ \vdots \\ u_{J-1} \\ u_J \end{array} \right\}_{J \times 1} \end{array} \right]^{n+1}_{2J \times 1} \\
 & = \left[\begin{array}{c} \left[\begin{array}{c|c} \overbrace{\begin{matrix} 0 & -b_1 & & \\ -A_2 & 0 & -B_2 & \\ \vdots & \vdots & \vdots & \\ -A_{J-1} & 0 & -B_{J-1} & \\ & -a_J & 0 & \end{matrix}}^{[C]_{J \times J}} & \overbrace{\begin{matrix} q_1 & -e_1 & & \\ -C_1 & Q_2 & -E_2 & \\ \vdots & \vdots & \vdots & \\ -C_{J-1} & Q_{J-1} & -E_{J-1} & \\ & -c_J & q_J & \end{matrix}}^{[D]_{J \times J}} \\ \hline \overbrace{\begin{matrix} p_1 & -i_1 & & \\ -G_2 & P_2 & -I_2 & \\ \vdots & \vdots & \vdots & \\ -G_{J-1} & P_{J-1} & -I_{J-1} & \\ & -g_J & p_J & \end{matrix}}^{[G]_{J \times J}} & \overbrace{\begin{matrix} l_1 & m_1 & & \\ K_2 & L_2 & M_2 & \\ \vdots & \vdots & \vdots & \\ K_{J-1} & L_{J-1} & M_{J-1} & \\ & k_J & l_J & \end{matrix}}^{[H]_{J \times J}} \end{array} \right]_{J \times 2J} \left[\begin{array}{c} \left\{ \begin{array}{c} v_1 \\ T_1 \\ T_2 \\ \vdots \\ T_{J-1} \\ T_J \end{array} \right\}_{J \times 1} \\ \left\{ \begin{array}{c} v_2 \\ u_1 \\ u_2 \\ \vdots \\ u_{J-1} \\ u_J \end{array} \right\}_{J \times 1} \end{array} \right]^{n+1}_{2J \times 1} + \left[\begin{array}{c} \left\{ \begin{array}{c} c_1 \\ f_1 \\ F_2 \\ \vdots \\ F_{J-1} \\ f_J \end{array} \right\}_{J \times 1} \\ \left\{ \begin{array}{c} s_1 \\ 0 \\ \vdots \\ 0 \\ s_J \end{array} \right\}_{J \times 1} \end{array} \right]_{J \times 1}
 \end{aligned}$$

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